A field study examining how workplace diversity impacts the recruitment of minority group members

Aaron D. Nichols[1*], Jordan Axt[2*], Evelyn Gosnell[3], Francesca Gino[4], and Dan Ariely[5]

[1] Questrom School of Business, Boston University, Boston, MA, USA

[2] McGill University, Montreal, QC, CA

[3] Irrational Labs, San Francisco, CA, USA

[4] Harvard Business School, Harvard University, Cambridge, MA, USA

[5] Center for Advanced Hindsight, Duke University, Durham, NC, USA

*** Corresponding authors:** Aaron D. Nichols (adn@bu.edu; https://orcid.org/0000-0003-2536-5850); Jordan Axt (jordan.axt@mcgill.ca; https://orcid.org/0000-0003-3944-8331).

Abstract

Increasing workplace diversity is a common goal among many employers, and one primary avenue for improving diversity is through greater recruitment of applicants from minority groups. Given prior research suggesting that minority applicants anticipate being treated with more respect and fairness in workplaces that are already more diverse, we will run a naturalistic field experiment that explores how subtle organizational diversity cues affect the behavior of applicants from both minority and majority groups. In the experiment, participants searching for employment will be exposed to a company with varying levels of racial and gender diversity. We will then investigate whether the decision to submit a job application is impacted by the presence or absence of certain forms of diversity, and if these diversity cues also influence overall applicant quality. Results—and even null results—can advance understanding of current issues related to organizational behavior, employee recruitment, and intergroup processes.

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Increasing workplace diversity is a popular but elusive goal for many employers. Over the past several decades, equal opportunity initiatives and new policies in business have encouraged employers to make concerted efforts to hire diverse personnel.[1] However, despite numerous regulations and new efforts in diversity recruitment, many organizations struggle to hire employees from minority backgrounds,[2,3] with even well-intentioned initiatives to improve diversity backfiring or creating adverse reactions among minority employees.[4,5] This lack of workplace diversity is particularly problematic within white-collar industries, where Black men and women are still underrepresented;[6] as recently as 2015, there were more Fortune 500 CEOs named “John” or “David” than there were women.[7,8] Why are these organizations unable to increase the diversity of their workforces? How can organizations encourage applications from talented employees that come from all walks of life? In this work, we investigate how workplace diversity cues affect the quality and quantity of job applications from both minority and majority groups.

One difficulty that organizations may have in creating a diverse workforce is in communicating to applicants that their organization is one where members of minority groups can anticipate being treated with respect and fairness. According to social identity theory,[9] individuals attend to important characteristics (i.e., age, gender, and race) to create identities and to differentiate themselves and others. Individuals often have multiple social identities[10-12] and the salience of these identities can influence behavior. For instance, individuals tend to seek out

environments that affirm their identity,[13] especially when they have previously experienced discrimination.[14]

Past work has shown that social identity is a key factor in determining individuals' organizational commitment and comfort.[13,15,16] Not only do employees use their organizational affiliations to form their social identity, they also experience more positive outcomes when these organizations reflect their existing social identity and values. For instance, racial minorities working in organizations with greater perceived commitment to diversity expressed a lower desire to leave their current positions.[17] Conversely, in contexts where minority applicants anticipate discrimination, they are more likely to alter their presentation as a potential employee and engage in "resume Whitening", such as through removing experience in organizations that could signal racial identity (e.g., National Society for Black Engineers).[18] To the extent that such positive or negative outcomes can be anticipated, applicants from minority groups should be more attracted to organizations where they are more similar to other employees.

A straightforward implication of this prior work is that organizations could communicate to applicants from minority groups that they would be entering an inclusive environment through front-facing materials, like company websites, that highlight existing organizational diversity. Indeed, prior work has found that people readily use such materials to form impressions of an organization. For instance, it was observed that when company-brochure images depicted a high level of minority representation, Black professionals were more trusting of the company regardless of the company's stated diversity philosophy.[19] Furthermore, Black professionals' increase in organizational trust was mediated by a decrease in the activation of social identity contingencies- that is, concerns about being marginalized because of one's racial identity. This

work highlights how workplace diversity cues are important to workers from underrepresented groups, as these cues can affect trust and social identity contingencies.

Similar findings have emerged in other studies. For example, one lab study found that presenting a fictional departmental website with images that displayed greater diversity in the student population (here, in terms of age), led student participants to think the department placed a greater value on diversity and that they would be treated with more respect.[20,21] Additionally, a field study among ethnic minorities in Belgium found that job advertisements were rated less positively when they stressed the importance of traits (e.g., integrity) that minority participants believed to be the topic of negative stereotypes that others held towards their own group.[22] In general, the research indicates people are sensitive to job materials that subtly indicate a potentially unwelcoming environment.

Together, these studies highlight correlational and experimental evidence that members of minority groups experience more positive outcomes in contexts with greater diversity, and that minority group members will use organizational materials like websites and job advertisements to infer the likelihood of feeling welcomed. Although this work is certainly informative, the literature has noticeably few field studies that combine the external validity of investigating real-world decisions with experimental manipulations that can identify sources of causality in behavior. Moreover, while there are several field studies that use experimental approaches to identify how employers evaluate applicants from minority groups,[18,23,24] fewer field studies investigate how applicants use information related to an employer's treatment of minority group members in order to make consequential employment decisions.

A field study on organizational diversity cues would be highly useful for two reasons. First, many prior studies finding that members of minority groups experience more positive

outcomes in more diverse workplaces are ultimately correlational, thereby leaving open the possibility that other factors are responsible for this relationship (e.g., organizations with greater diversity are also more likely to invest in forms of employee wellness or provide better benefits). Second, the existing experimental evidence showing the impact of diversity cues has needed to rely on secondary outcomes (e.g., anticipated respect[20]), all while completed in an environment where participants know their responses will be evaluated. Indeed, the literature on workplace diversity cues is limited in scope and external validity. The prior work has relied on outcomes like self-reported organizational attractiveness,[19,22,25-31] and anticipated trust,[19,29] rather than a more externally valid measure like willingness to apply to join an organization. In sum, while research indicates subtle details can induce social identity threats,[32-34] decrease organizational trust and comfort,[19,29] and weaken job appeal,[19,22,25-31] the literature lacks direct evidence that organizational cues can affect actual applicant behavior in an ecologically valid context.

We aim to build upon existing research and address this prominent gap in the literature by conducting a naturalistic field study testing the effect of diversity cues, conveyed through online images of an organization's members, on the quality and quantity of applicants from both minority and majority groups. We study the recruitment of female and non-White applicants, and do so in a context where women and people of color are stigmatized and biases are prominent: technology.[35-37]

To do so, we created a realistic website for a hypothetical technology startup company with several job openings. Visitors were recruited through online advertising and were randomly assigned to view images of the company's workforce that were either high or low in gender diversity, racial diversity, or both. Using a 2 X 2 between-subjects experimental design, we manipulated the presence or absence of racial and gender diversity, such that images of the

organization's employees were either: 1) all White men 2) White men and White women, 3) White men and non-White men, or 4) both White and non-White men and women. For comparison purposes, we operationalized the all-White male condition as our positive control. We investigated whether this manipulation of organizational diversity cues impacted the behavior (i.e., the decision to apply for a job) of potential applicants. In order to maximize statistical power and minimize threats to statistical validity, we investigated workplace diversity effects across stigmatized and non-stigmatized groups based on race (non-White vs. White) and gender (female vs. male). Specifically, we examined how such subtle organizational diversity cues affected the quantity and quality of applications that were submitted from non-White men, non-White women, White men, and White women.

In addition to using a realistic and consequential outcome like the decision to submit a job application, our design allows for tests concerning how members of minority and majority groups respond to different signals of organizational diversity. For one, this study can investigate the extent to which applicant behavior is impacted by the presence of organizational diversity cues that promotes members of other minority groups (e.g., how non-White men react to an organization that seems willing to hire women but not racial minorities). Second, this work can explore the behavior of members of majority groups (i.e., White men), which might add nuance to prior work that has observed conflicting evidence concerning the degree to which majority group members react to organizational diversity cues.[25,27-29,38]

Taken together, this work can both advance understanding on issues concerning employee recruitment and intergroup relations, while also providing an externally valid investigation of organizational diversity cues and their role in the recruitment of minority group members. Below, we review the prior evidence that informs possible study outcomes.

First, one clear prediction from past research is that applicants from minority groups will have more positive evaluations of organizations that employ members who share their same identity, leading to a greater likelihood of submitting a job application. Previous work consistently finds that minority group members respond positively to seeing others with similar identities in organizational materials. In one example, Black female undergraduates were presented with a picture of a fictional school's science department, and then read a profile of a faculty member that was either a Black or White male or female.[39] Results found that participants indicated greater anticipated belonging and trust at the school when presented with a Black faculty member (either male or female). Similarly, Black female participants evaluated a hypothetical technology company more positively when the company's materials included the profile of either a current male or female Black employee.[40] Moreover, Latina women rated a hypothetical STEM company more positively when the company displayed the profile of a current Latino or Latina employee.[41]

These lab studies are complimented by correlational research that finds greater job satisfaction among minority employees working in settings where they share greater similarity with those around them. For instance, employees who were racial/ethnic minorities reported lower levels of perceived discrimination at work when their supervisor shared their demographic background.[42] Conversely, racial minority students working in study groups with greater ethnic dissimilarity to themselves reported lower feelings of belonging and reduced organizational attachment.[43]

To the extent that participants from minority groups can anticipate the positive consequences of working in an organization where other coworkers have the same identity, one hypothesis is that workplace diversity cues will affect participants' preferences and job

application behavior. Specifically, members of minority groups should be more likely to submit a job application when viewing a company whose members show greater similarity to their own stigmatized identity or identities (see Hypotheses 1,3 and 5 in Table 1). In this case, results would anticipate White women applying more to organizations that are shown to have greater gender diversity and Non-White men applying more to organizations that are shown to have greater racial diversity. A small exception concerns non-White women, who are also expected to apply more to organizations showing greater racial (but not gender) diversity, given prior findings of Black and Latina female participants showing greater organizational attractiveness when materials highlighted a female or male employee of the same racial/ethnic identity.[40,41]

However, another series of possible outcomes emerges from more recent work concerning how members of minority groups will be impacted by organizational diversity cues that promote members with differing under-represented identities.

While there are relatively straightforward predictions concerning how job applicants from minority group will evaluate organizations whose members share their own race or gender, there is more mixed evidence on how such applicants will evaluate organizations whose workforces signal a different kind of diversity, one that promotes other under-represented groups (e.g., White women viewing an organization that has racial but not gender diversity). Some research suggests that the impact of diversity may be rather narrow, as minority group members' evaluations of organizational diversity are overly reliant on the presence or absence of group members who share one's own stigmatized identity. Moreover, the impact of diversity messaging may also be specific to certain marginalized identities; for example, Black female participants' evaluation of a hypothetical company was affected by materials promoting racial but not gender diversity.[39]

Indeed, people appear to be biased in their assessments of diversity. For instance, Black and Asian participants gave lower ratings of diversity when a company had minority employees that differed from participants' own race; that is, Asian participants thought there was lower diversity at a company whose only racial minority employees were Black compared to a company whose only racial minority employees were Asian.[44] Applying these findings to our own design, we might anticipate that perceptions of organizational diversity, a key factor in determining whether members of minority groups will apply for a job, will be impacted by the degree to which applicants see their own racial or gender identities reflected in the current employees (e.g., White women should view the all-White company that has gender diversity as more diverse, and therefore more appealing, than the all-male company that has racial diversity).

At the same time, such results are in contrast with work finding "spillover" biases in diversity judgments. In one example, participants evaluating groups of faces containing equal numbers of males versus females across groups- but differing numbers of racial minority members- provided higher ratings of gender diversity to those groups with more racial diversity.[45] Related effects have emerged in work on "identity safety cue transfers", where people from minority groups use commitments to a specific form of diversity in order to infer broader egalitarian values. In one example, White female participants' perceptions of fairness towards women and anticipated inclusion was lower in a company that implemented a general training program that did not mention race or gender, but these perceptions were ameliorated when the company implemented a training program that either supported women or racial minorities,[46] with similar effects emerging among male Black and Latino participants. Conversely, female participants anticipated greater sexism from an evaluator displaying racially biased behavior, and male racial minority participants anticipated greater racial bias from an

evaluator engaging in sexist behavior.[47] This work aligns with separate findings that although some groups are more associated the concept of diversity than others,[48-50] researchers have observed a sizeable overlap in the shared identity of minorities, especially amongst people of color.[51-53] Similarly, individuals with at least one stigmatized identity (i.e., White women or Black men) may be more sensitive to issues of privilege, and respond more positively to all forms of organizational diversity, than individuals with no stigmatized identities (i.e., White men[12]).

These findings lead to diverging predictions on how the behavior of minority group members in this field study will be impacted by differing forms of organizational diversity. If willingness to submit an application is primarily tied to seeing an organization hire people sharing one's own racial or gender identity,[44] female White participants should be impacted by seeing organizations with gender but not racial diversity, and male non-White participants should be impacted by seeing organizations with racial but not gender diversity. However, if the decision to submit an application is sensitive to these identity safety transfer cues, then any form of diversity should impact members from minority groups.[46]

This second perspective would predict that White women, non-White women, and non-White men will apply more to organizations that promote any form of racial or gender diversity. That is, analyses should produce a reliable interaction between the racial and gender diversity manipulations, such that rates of applying are only lower in the organization made entirely of White men that lacks both racial and gender diversity (see Hypotheses 2, 4 and 6 in Table 1).

A final question concerns how members of majority groups (here, White men) will be impacted by varying forms of organizational diversity. Prior studies have found that, compared to minority group members, majority group members differ in the cues and information they use

to assess organizational diversity.[54-56] Indeed, majority group members have been found to be relatively insensitive to organizational diversity cues.[27-29] For instance, whereas the manipulation that introduced a generic, gender-specific, or race-specific training program impacted beliefs about being treated fairly among female and racial minority participants, White male participants showed no effect across conditions.[46] Similarly, White participants' intentions to pursue a job were unaffected by messages indicating that the organization had instituted a mentoring program that was limited to Black employees.[57]

However, separate work has found that members of high-status or majority groups can feel threatened by signs of increased diversity, as Whites implicitly and explicitly associate the concept of "multiculturalism" with that of exclusion (versus inclusion).[58] In the job seeking context, past studies have found that White participants applying for a hypothetical job reported more concerns about receiving unfair treatment or less organizational attractiveness when the organization mentioned a commitment to diversity,[59,60] results that would align with more general findings of greater threat experienced by Whites shown information about increased racial diversity in the US population.[61,62] Moreover, past work has found that White participants found an organization to be less attractive when the organization's recruitment website features a greater proportion of testimonials from minority employees.[25] Finally, correlational research among actual employees has also highlighted lower willingness to embrace organizational diversity among majority group members; one large-scale survey of over 3,000 employees revealed that, compared to White women and racial/ethnic minorities, White men reported lower comfort with and perceived value in organizational diversity.[38]

Such reactions towards the concepts of diversity could make it less appealing for members of majority groups to submit applications to organizations that appear to be embracing

diversity. From this perspective, one hypothesis is that results would produce an adverse reaction among White men to the concept of diversity, finding that White men will apply less to organizations that have racial diversity (see Hypothesis 7 in Table 1) or both racial and gender diversity (i.e., main effects of both the racial and gender diversity manipulation; see Hypothesis 8 in Table 1).

Compared to applicants with stigmatized identities, it is less clear how members of majority groups (here, White men) will react to manipulations showing varying forms of organizational diversity. To that end, there is even the possibility that White men will respond positively to greater organizational diversity, given the prior finding that materials promoting organization diversity resulted in more positive perceptions from both minority and majority group members.[21] This work then provides an additional test, using a naturalistic and externally valid setting, of how majority group members respond to information related to organizational diversity.

Interestingly, research on diversity and employee recruitment has largely ignored the role of applicant quality.[63,64] However, it may be the case that reactions to diversity cues vary depending on applicant quality. For instance, previous models suggest that targeted recruitment based solely on demographics may lead to a greater number of applications submitted by less qualified minority jobseekers, thereby weakening the overall quality of applications from minority jobseekers.[63]

The opposite effect may emerge for more qualified applicants with stigmatized identities, as a prior field study found that organizations promoting diversity through an equal employment opportunity (EEO) statement received fewer applications from more qualified minority applicants. Follow-up surveys suggested that this phenomenon was due to “tokenism” concerns;

highly qualified minority applicants wanted to avoid settings that were more likely to hire for symbolic purposes rather than for merit.[65] To the extent that our own manipulations create similar tokenism concerns, it is possible that they may reduce motivation to apply among more qualified minority applicants. In the present research, it may then be the case that signaling a commitment to diversity by increasing the diversity of the organization's workforce will increase the number of applications from less qualified stigmatized applicants and decrease the number of applications from more qualified stigmatized applicants. As a result, one hypothesis from prior research is that increased diversity in recruitment materials will result in lower applicant quality from applicants with stigmatized backgrounds who see employees sharing their own racial or gender identity. For example, this perspective would anticipate that non-White men will show lower levels of applicant quality when viewing organizations that have any racial diversity (i.e., a main effect of racial diversity; see Hypotheses 9-11 in Table 1). Accordingly, if White male applicants view diversity as a potential threat that deters less qualified applicants, average quality of White male applicants may increase when organizations are shown with greater racial or gender diversity (see Hypothesis 12 in Table 1).

This process may produce a somewhat ironic effect of greater diversity in organizational materials leading to weaker overall applicant quality from members of stigmatized groups. We test this possibly directly by hiring professionals with human resources experience to rate the quality of applicants from both stigmatized and non-stigmatized groups. We recognize that evaluations from individuals with experience in human resources are still subjective, so such analyses cannot speak to any truly objective measure of applicant quality. However, this approach has the dual benefit of both capitalizing on our raters' prior expertise in evaluating job

applicants and closely mirroring the method by which many organizations screen potential employees.

By observing application behavior in a realistic setting, this work will advance understanding of how various signals related to organizational diversity affect the demographic makeup, quality, and overall size of an applicant pool. In particular, this study investigates: 1) if prior lab findings that members of minority groups show greater interest in organizations that highlight demographic similarity among current employees[39] can be reproduced in field settings, 2) if members of minority groups will be impacted by subtle signals of organizational diversity, even when such diversity does not reflect their own identity,[46] 3) if members of majority groups will show a negative reaction (i.e., lower likelihood of submitting an application) to greater organizational diversity,[54] and 4) how the overall level of applicant quality is influenced by organizational diversity cues.[65]

Finally, it is important to consider the appropriate interpretation of possible null effects. If null effects are observed, it would be inaccurate to conclude that organizational diversity has no effect on applicant behavior or quality. Rather, null effects should only be interpreted as a lack of evidence that this particular manipulation of cues to organizational diversity impacted job applicants. Given the field design of this work, there are various unobservable factors that prevent more specific conclusions regarding null results.

Results

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Discussion

[This is a placeholder section. We will address study limitations and future directions as previously outlined in our responses to the editorial team]

Methods

Ethics

This study has received approval from the Duke University IRB (Protocol # 2018-0264). Given the field setting of the study, we obtained waivers for consent, compensation, and debriefing. Identifiable data will be stored on password-protected computers, while de-identified data and analysis code will be shared publicly on the Open Science Framework (OSF) at https://osf.io/vaq2g/?view_only=5de809528aae49d8887170b63dacbee6. See Table 1 for details and links for each resource available in the online supplement.

Procedure

Participant recruitment. We will recruit participants online by posting up to nine jobs on LinkedIn and promoting them with a daily budget between $15-$35. The nine jobs will be at a company called "Foodable", which is a "Silicon Valley based company on a mission to redefine how people purchase and consume food." The job posts will prompt potential applicants to apply directly on the company website and include that we have created for the company (https://foodabletech.com/). This website presents basic background information about the company, displays an image showing current employees, and lists up to nine job openings. See Appendix A for website content.

Experimental manipulation. After clicking the link to the Foodable web page, participants will be randomly assigned to one of four experimental conditions, following a 2 (Racial Diversity: Yes vs. No) by 2 (Gender Diversity: Yes vs. No) design. The experimental manipulation of organizational diversity will be conveyed through an image shown on the 'Team' heading of the website, as well as on through an image on the bottom of the page. Participants will be randomized to experimental condition after opening the website link.

Specifically, participants will be randomized to view the version of the website with the least number of assigned visitors, ensuring comparable numbers of participants across conditions.

In order to minimize the risk of exposure to multiple website conditions, condition assignment will be maintained utilizing the participant's HTTP cookies and their IP address. In the background of their browsers, participants are sent HTTP cookies by accessing the website. Critically, the website links a participants' HTTP cookies and IP address to their condition. This two-factor authentication ensures that every new visitor that has either a) the same HTTP cookies and/or b) the same IP address will see the version of the website they saw at first visit. If the participant disables HTTP cookies or clears their cookies between visits, they will view the website condition assigned to their IP address. Ultimately, maintaining assignment using IP addresses helps prevent participants from detecting condition within networks (i.e., across devices at home, at the office, at the library), while also maintaining assignment with HTTP cookies helps prevent participants from detecting condition between networks when the same device-browser combination is used (i.e., accessing the website from Google Chrome on a mobile phone at home and then later at work). For illustrative examples and more details about condition assignment procedures, see the link to "Condition.Assignment.Maintenance" in Table 1.

Each condition will show 16 team members presented with headshots and names. See Appendix B for stimuli from each experimental condition. In the No Racial Diversity, No Gender Diversity condition, the team will consist entirely of White males. In the No Racial Diversity, Gender Diversity condition, the team will consist entirely of White people, but will have eight males and eight females. In the Racial Diversity, No Gender Diversity condition, the team will consist entirely of males, with eight White males and eight males from racial minority

backgrounds (three Black, two East Asian, two South Asian, and one Hispanic). Finally, in the Racial Diversity, Gender Diversity condition, the team will have four members for each combination of male versus female and racial minority versus White. To make this racial information more salient, racial minority team members have been given common surnames commonly associated with that racial/ethnic group (e.g., Garcia, Nguyen, Gupta). Similarly, team members will be shown first names that are typically associated with male or female genders (e.g., Michael, Jake, Jason vs. Danielle, Dawn, Josephine).

Application submission. Across conditions, the website will list up to nine job openings: senior product manager, marketing communications manager, embedded systems architect, account manager, web developer, copy writer, business analyst, recruiter, and full stack engineer. These positions were chosen due to their relevance within a technology company. In an effort to increase sample size, the positions include a range of possible skillsets and will be posted online as needed. The manuscript will report analyses on the data collapsed across positions while all position-level data will be available online. Given the rapid growth many start-ups experience, adding more job postings may have the additional benefit of increasing the believability of "Foodable". Notably, all potential applicants must scroll past the 'Team' section of the website before being accessing information about job openings. After clicking on each job title, in-depth information about each position will be displayed (responsibilities, qualifications, benefits). To write the job descriptions, we consulted with people who had experience working in each field. See Appendix C for text from each position.

When viewing each position, participants can select the 'Apply Now' button to begin the application process. The application is completed entirely online and does not require participants to upload a resume. Participants are informed that they may be asked to provide a

resume at a later time. See Appendix D for a sample application portal for the stack engineer position. After first providing basic contact information, participants report their educational background, educational performance (i.e., GPA), and performance in math and science classes. Participants then report any awards or honors they have received, a self-assessment of their performance as a student, as well their total years of work experience. Participants will then report their work background, information about their previous job titles, responsibilities, length of prior employment, and a self-assessment of effectiveness for their previous jobs and internships. Next, participants provide ratings of their proficiency in various job-related skills (e.g., JavaScript programming), indicate their willingness to review some background information about the company, and list their available times for a possible follow-up interview. Finally, participants will indicate how they heard about the job advertisement (i.e., through an online advertisement, an online search, or a referral) before finally reporting their gender and race before submitting their application. For analytic purposes, we will collapse anyone who reports race as not White into the “non-White” category, and we will exclude participants who report gender as “Other”, though all participants will be available in our online dataset.

Applicant evaluation. To develop a system of evaluating applicants, we worked with professionals who had prior experience in human resources to create an evaluation rubric (see Appendix E). The rubric was designed to provide an overall assessment of each applicant that could be completed in less than 10 minutes.

After data collection is finished, we will hire two new coders that also have prior experience in human resources. Coders will be trained together and asked to review groups of sample applications from a pilot study until they achieve acceptable reliability ($i > .60$). Specifically, we will collect applications from 30 MBA students to train our coders. The two

coders will assess application quality for 20 applications randomly selected from our training sample. If inter-rater reliability (*i*) is less than .60, we will have the coders meet to review their 20 assessments and discuss the alignment of grading strategies. Afterwards, the coders will review the remaining 10 applications and inter-rater reliability will be assessed for these 10 applications. If inter-rater reliability remains below .60 after this procedure, we will collect more sample applications and repeat the training process.

Once evaluators achieve this level of inter-rater reliability, each of the actual applicants will be assigned to one of the two coders. Coders will be blind to experimental condition and will be operating under the assumption that some of these applicants may actually be hired for a position. Using a 1= Not acceptable to 5= Highly acceptable scale, coders will rate each applicant on the following dimensions: interpersonal skills; education/knowledge; knowledge and skills in research; leadership and collegiality. These four ratings will then be summed to create a total qualification score, which we will use as our dependent variable when analyzing applicant quality.

Importantly, coders will evaluate de-identified applications (i.e., removing names, email addresses) that are also blinded for race and gender. Application de-identification helps protect participants' personal information, while application blinding helps ensure any biases specific to raters are spread equally across applicants, regardless of race or gender. However, this blinding of applicant demographic information may be compromised if the applicant attended a historically Black college or university (HBCU) or an all-women's college. To address this possible issue, an author will first screen all eligible applications for whether the applicant attended an HBCU or an all-women's college. If there are no such cases, there will be no changes to the evaluation procedure. However, if there are applicants who went to an all-

women's college or HBCU, then that school will be replaced with a school of equal or comparable ranking according to the latest college rankings from the US World and News Report.

Participants

Our sample size calculations are complicated by the anticipated unequal distribution in applications received from White versus non-White employees. As LinkedIn does not publish information regarding the gender or racial breakdown of their users, we relied on estimates from a recent report concerning the demographic composition of workers in the tech industry. A 2016 report from 177 tech companies (see Table 2) indicated the following distributions of gender/racial identities: 41.3% White men, 16.2% White women, 28.5% non-White men, 14.0% non-White women.[66] Using this distribution as an estimate for our own sample size, we ran a series of power analysis simulations separately for White men, non-White men, White women and non-White women for our focal analysis concerning a logistic regression on likelihood of being a member of the applicant pool based on organizational gender and racial diversity (see Table 2).

Given our competing hypotheses of whether to expect main effects of either racial or gender diversity or interactions between the two factors, we provide power calculations for both outcomes. For non-White men, a sample size of 720 total eligible applicants (i.e., from both White and non-White men and women), would provide 95.6% power to detect a small-to-medium main effect (Odds Ratio = 1.72, Cohen's d = .30) of racial diversity.[67] For non-White women, a sample of 1036 participants would provide 95.9% power to detect a main effect of racial diversity of the same magnitude. For White women, a sample size of 908 would provide 97.5% power to detect a gender diversity main effect of OR = 1.72. Finally, for White men, a

sample size of 944 would achieve 96.1% power for detecting a main effect of racial diversity of OR = 0.58 (Cohen's d = .30), since racial diversity may suppress applications among White men.

The largest number from the above analyses is N = 1036 for the non-White woman analysis. As a result, we consider 1036 applicants to be our absolute minimum sample size. This sample will be able adequately powered to detect theoretically-relevant effects of racial and gender diversity on applicant recruitment, and null results will be informative for ruling out possible effects at reasonable effect size estimates (Cohen's d = .30).

We consider this effect size to be the lower limit for testing a theoretically and practically meaningful intervention. For one, this intervention is aimed at impacting only the initial level of the hiring process (i.e., the size and makeup of the applicant pool). Job applicants are exposed to many more levels of evaluation, and final hiring decisions are typically based on in-depth evaluations, like interviews, of only a handful of applicants. As a result, any intervention concerning the first step of the hiring process (i.e., submitting an application) would have to be at least moderate in size in order to effectively carry over into meaningful changes in later stages of hiring. Second, the intervention would need to have at least a moderate effect size in order to convince organizations to adopt similar practices; many organizations may not be willing to change their employee recruiting processes if the results of this study produced only negligible changes to the applicant pool. Our power analyses are also sensitive to detect effects of similar magnitude in the research literature. For example, Black female participants reported more anticipated belonging in fictional schools that presented Black versus White faculty profiles (d = .39),[39] with a comparable effect size emerging among Latino participants seeing a profile of a Latino employee (d = .39).[41]

However, while we consider 1036 participants to be an absolute minimum, we will seek to collect up to 1524 participants. This number is based on a series of follow-up simulations concerning a possible interaction effect between our racial and gender diversity manipulations (specifically, such that the impact of diversity on likelihood of submitting an application would be equal across organizations that had any form of diversity). Achieving approximately 95% power for detecting a medium interaction effect (OR = 2.07 or OR = 0.48, Cohen's d = .40), ranged between 1496 and 1524 depending on demographic group. The online supplement (see Table 2) has syntax for running each sample size calculation. As a result, while 1036 is our minimum sample size, we will attempt to continue collecting data until we reach a sample size of N = 1524. The obtainable sample size is hard to estimate given the field study design.

The N = 1036 target sample size will also provide sufficient power for other planned analyses. Our second proposed analysis concerns applicant quality. To do so, we will hire trained coders to rate applicants. Using the above sample size estimates, coding the 1036 expected applicants would be prohibitively expensive. Instead, we plan to have coders rate a subset of the total applicants. To maximize statistical power, we will have coders rate all of the female and non-White male applicants. As a result, 95% power will be achieved for detecting a main effect of either racial or gender diversity of d = .60 among non-White women, d = .57 among White women, and d = .42 among non-White men. We will then have coders rate a random sample of 75 White men per experimental condition, which would also allow for 95% at detecting a main effect of d = 42 for either racial or gender diversity.

Finally, for our third analysis concerning whether the experimental manipulations impacted overall rates of submitting an application from website visitors. Assuming a modest baseline of 3% of website visitors submit an application, the sample size would provide over

95% power to detect an odds ratio effect size as small as OR = 1.42 for a main effect of either racial or gender diversity or an interaction between the two factors, an effect size that is below the benchmark of a "small" effect (OR = 1.68).[67]

Data and Materials Availability

We will share all materials, including source code for creating the websites, on the Open Science Framework. Our primary analyses will focus on how experimental condition impacts the quantity, quality and diversity of applicants, and we will share the necessary data and syntax needed to reproduce those analyses. Sample data and syntax can be viewed at https://osf.io/vaq2g/?view_only=5de809528aae49d8887170b63dacbee6. However, due to confidentiality concerns, we will not be storing certain variables (e.g., names and email addresses) nor will we be able to make certain variables publicly available (e.g., undergraduate institutions, names of prior companies where employed).

Data Treatment and Analysis

Exclusions

We do not anticipate excluding any participants from this study. We may remove certain participants who we believe are submitting fake applications (e.g., providing clearly fake email addresses, listing over 100 years of work experience, reporting being the CEO of Apple). Furthermore, we will exclude participants to control for word-of-mouth exposure, attention, and manipulation-avoidance. Specifically, we will remove participants that indicated they heard about the job through referral. We will also remove participants that spend less than thirty-seconds on the "Foodable" webpage before accessing the application. Finally, we will exclude anyone who circumvents the manipulation and accesses the application page without seeing the Foodable webpage. It is hard to anticipate how many of these cases there will be, if any.

However, if such cases arise, we will conduct our primary analyses on both the final sample and the original sample, and results from both samples will be reported in an online supplement. If our exclusions do not substantively change the results, only the analyses on the final sample will be reported in the manuscript. Importantly, these exclusion criteria will be applied within the measurement period (from the participant's first visit to their subsequent application submission).

In addition to the circumstances outlined above, we recognize that having multiple visitors with the same IP address could complicate our conclusions. To explore this issue, we will also re-run our analyses excluding cases where a new visit is logged from a duplicate IP address after that IP address has already been associated with a submitted application; this process would exclude any new applicant that comes from an identical IP address. Although it is unlikely to occur often, it would be reasonable for a participant with a variety of skills to apply to multiple positions. Therefore, we will only exclude such participants from the final sample if our results are substantively different after duplicate IP addresses are excluded.

Data Quality Checks and Transformations

We do not anticipate having to make any transformations on our data. However, there are two potential concerns regarding the validity of our data that we will investigate before starting our primary analyses. One concern is that participants' self-ratings of their skills will have ceiling effects due to desires to create a strong impression on potential employers. To further examine this issue, we ran a pilot study among MBA students (N = 24), and asked them to complete the online application. When evaluating their own skillsets, only 9% (11/122) of responses were the highest possible value, and no single skill received more than 14% of responses being the highest value. We expect similar distributions in our field sample, but will

run a preliminary analysis investigating what percentage of the sample report having the highest level of a certain skill. If this percentage is high enough to introduce possible ceiling effects (specifically, greater than 40%), we will remove that variable from the data given to the human resource coders.

Planned Analyses

We anticipate running the following analyses to address our primary research questions. Note, the threshold for determining the significance of the analyses below will be set at $\alpha = 0.05$ and all analyses will be two-tailed tests.

Impact of organizational diversity on diversity of applicant pool. We will first analyze how our experimental manipulations impact the likelihood of someone submitting an application. Specifically, we will run four binary logistic regressions. The first regression (Analysis 1, see Table 1) will predict the likelihood of an applicant being a White women from the gender diversity condition (1= Yes, 0= No), racial diversity condition (1=Yes, 0=No), and their interaction, controlling for position type (using a series of dummy-coded variables). This analysis will be repeated three times (Analyses 2-4, see Table 1), using the outcome variable of the likelihood of the applicant being a White female, a non-White male, or a non-White female, respectively.

Given that applicants access the company's website using the exact same advertisements, we can assume that there will be no systematic differences across conditions in terms of the demographic makeup of who is exposed to each condition (e.g., there is no reason to believe that there will be systematic differences in the number of White men who view each condition). As a result, these analyses can identify if our manipulations of organizational diversity cues made both minority and majority group members were more or less likely to submit an application.

Impact of organizational diversity on applicant quality. Next, we will investigate the effects of our experimental manipulation on applicant quality. More specifically, we will analyze how our manipulations of organizational diversity cues could also impact which applicants from majority and minority groups decide to apply. For instance, if individuals from minority groups anticipate unfair treatment in organizations that lack diversity, this may lead to only the most qualified members submitting applications.[63] This process could lead to a paradoxical effect where conditions lacking organizational diversity have fewer applicants from minority groups, but those minority group members are actually of greater overall quality than those in conditions that saw higher levels of diversity. A similar effect could emerge for majority group members (i.e., White men); if they view diversity as a threat, only the most qualified applicants will apply to positions in more diverse organizations.

To investigate this question, we will also run a series of 2 (Racial Diversity: Yes vs. No) by 2 (Gender Diversity: Yes vs. No) between-subjects ANOVAs, with overall applicant score (as provided by our coders) as the dependent variable. We will run these analyses four times, once separately for 1) White female participants, 2) non-White male participants, 3) non-White female participants, and 4) White male participants (Analyses 5-8, see Table 1).

Impact of organizational diversity on number of applications started and completed. A final question of interest concerns whether any of the experimental manipulations simply resulted in more started or submitted applications in general. If greater racial and gender diversity is equally appealing to both majority and minority group applicants[21], there may be no impact on the percentage of applicants that come from minority versus majority groups, or on the average quality of applicant. However, such an effect would be evident on the general likelihood of participants submitting or starting an application at all (see Hypotheses 13-14 in Table 1).

To investigate this question, we will first run a binary logistic regression, predicting for each person who visits the website whether or not they submit an application (1= Yes, 0=No) from gender diversity condition (1= Yes, 0= No), racial diversity condition (1=Yes, 0=No), and their interaction (Analysis 9, see Table 1).

Next, we will run a parallel analysis (Analysis 10, see Table 1) predicting whether or not each visitor to the website *begins* an application. We believe this analysis is informative because it would suggest that our experimental manipulations are effective at raising interest in working for the organization among potential applicants. For instance, if we see effects on the number of applications started but not on the number of applications submitted, it would indicate that our manipulations heighten interest among potential applicants, but that our application procedure was overly difficult or long, meaning this initial interest could not be translated into actual application submissions. If so, we may want to run a follow-up study that has a more straightforward submission process (e.g., only asking applicants to upload a resume). To supplement this analysis, we also plan to run an exploratory analysis investigating whether our experimental manipulations impact how far applicants who do not submit an application progress in the application process (e.g., number of questions answered, number of pages viewed).

Given our experimental design, we cannot investigate whether any such effects are moderated by participant's own race or gender, as such information is only reported during the application process, meaning we lack demographic information among those who decide not to apply for a position. However, as mentioned earlier, all participants see the same advertisement text when first deciding to visit the company's website, so there should be no systematic bias in the number of participants from minority or majority groups who view each condition. As a result, if certain conditions are more likely to elicit submitted applications for specific participant

identities, that effect would emerge in our primary analysis described above. For example, given our recruitment procedures, we can assume there will be no systematic differences in how many White male participants are exposed to each experiment condition, so if certain conditions were particularly attractive to White males, that effect would emerge when looking at whether White males constituted a larger percentage of the applicant pool across conditions.

Robustness check. At the conclusion of these analyses, we will review the extent to which our conclusions do or do not change when alternative alpha levels (.01, .005, and .001) are used. We will report these robustness checks in the supplement and, if applicable, we will report in the manuscript which conclusions are altered when more stringent alpha levels are used.

Equivalence tests. Finally, we will run a series of equivalence tests[68] using the TOSTER R package[69]. For each hypothesis, equivalence tests will use the effect sizes listed in Table 1 to determine upper and lower equivalence bounds. These tests will assess if results there is support for the conclusion that any true effect is close enough to zero for practical purposes. For ease of comparison across tests and interpretability, each effect size will be converted into a correlation coefficient *r*. Equivalence tests will be run only when relevant results produce null effects. Table 2 provides a link to a sample analysis script that can be run for each of the relevant terms corresponding to Hypotheses 1-14.

Anticipated Timeline

We have already developed the study materials and form for evaluating applicants. We anticipate it will take six months to collect enough applications to meet our target sample size. Following that, we anticipate another three months for our human resource coders to evaluate the applicants. Finally, data analysis and write-up will require another two months. In total, we anticipate eleven months between the start of data collection and submitting a final report.

Protocol registration

The Stage 1 protocol for this Registered Report was accepted in principle on [TBD]. The protocol, as accepted by Nature Human Behaviour, can be found at [TBD].

Data availability

De-identified data and materials are available publicly on the Open Science Framework (OSF) at https://osf.io/vaq2g/?view_only=5de809528aae49d8887170b63dacbee6. See Table 1 for details and links for each resource available online.

Code availability

Code and syntax for analyses are available on the OSF at https://osf.io/vaq2g/?view_only=5de809528aae49d8887170b63dacbee6. See Table 1 for details and links for each resource available online.

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Author contributions

Authors ADN, FG, and DA conceptualized the research project. All authors contributed to study design. JA and ADN drafted the manuscript with remaining authors providing edits. JA formulated and executed the analysis plan. ADN, EG, and JA coordinated data collection. EG directed the technical development of our website stimuli. All authors provided key insights and feedback during the revision of this manuscript.

Competing interests

The authors declare no competing interests.

Table 1. Hypotheses Table

Question	Hypothesis	Sampling plan	Analysis plan*	Interpretation of different outcomes
Question 1: How do members of a stigmatized group (White women) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 1: White women will apply more to organizations that are presented as having any gender diversity.	Data collection will continue until a minimum of 1036 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 1.72 (*d* = .30) main effect of gender or racial diversity.	Analysis 1: Logistic regression predicting the likelihood of applicant being a White woman from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 29-38.	1) A main effect of gender diversity, such that application rates are higher in conditions with gender diversity, would suggest White female applicants respond positively to seeing ingroup gender diversity reflected in organizational materials. 2) A main effect of gender diversity and a main effect of racial diversity, such that racial and gender diversity is associated with higher application rates, would suggest White female applicants respond positively to seeing both ingroup gender and outgroup racial diversity reflected in organizational materials. 3) Null results showing neither main effect of

				gender nor racial diversity would suggest no evidence that our manipulations impacted application rates of White women.
Question 1: How do members of a stigmatized group (White women) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 2: White women will apply more to organizations that are presented as having any level of either racial or gender diversity.	Data collection will continue until a minimum of 1524 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 2.07 (d = .40) interaction between racial and gender diversity.	Analysis 1: Logistic regression predicting the likelihood of applicant being a White woman from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 29-38.	1) See above for interpretations of main effects. 2) An interaction between gender and racial diversity would suggest that, for White females, the impact of the gender diversity manipulation depends on the presence or absence of the racial diversity manipulation. 3) Null results showing no gender by racial interaction would suggest that, for White females, there was no evidence that the presence of the racial diversity manipulation moderated the impact of the gender diversity manipulation (and vice versa).

Question 2: How do members of a stigmatized group (non-White men) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 3: Non-White men will apply more to organizations that are presented as having any racial diversity.	Data collection will continue until a minimum of 1036 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 1.72 (*d* = .30) main effect of racial diversity.	Analysis 2: Logistic regression predicting the likelihood of applicant being a non-White man from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 40-49.	1) A main effect of racial diversity, such that application rates are higher in conditions with racial diversity, would suggest non-White male applicants respond positively to seeing ingroup racial diversity reflected in organizational materials. 2) A main effect of gender diversity and a main effect of racial diversity, such that racial and gender diversity is associated with higher application rates, would suggest non-White male applicants respond positively to seeing both ingroup racial and outgroup gender diversity reflected in organizational materials. 3) Null results showing neither main effect of gender nor racial diversity would suggest no evidence that our manipulations impacted application rates of non-White men.

Question 2: How do members of a stigmatized group (non-White men) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 4: Non-White men will apply more to organizations that are presented as having any level of either racial or gender diversity.	Data collection will continue until a minimum of 1524 eligible applications are submitted, is expected to provide greater than 95% power to detect an OR = 2.07 (*d* = .40) interaction between racial and gender diversity.	Analysis 2: Logistic regression predicting the likelihood of applicant being a non-White man from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 40-49.	1) See above for interpretations of main effects. 2) An interaction between gender and racial diversity would suggest that, for non-White males, the impact of the racial diversity manipulation depends on the presence or absence of the gender diversity manipulation. 3) Null results showing no gender by racial interaction would suggest that, for non-White males, there was no evidence that the presence of the racial diversity manipulation moderated the impact of the gender diversity manipulation (and vice versa).
Question 3: How do members of a stigmatized group (non-White women) respond to recruitment materials displaying	Hypothesis 5: Non-White women will apply more to organizations that are presented as	Data collection will continue until a minimum of 1036 eligible applications are submitted, which is expected to provide greater than 95%	Analysis 3: Logistic regression predicting the likelihood of applicant being a non-White woman from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded	1) A main effect of racial diversity, such that application rates are higher in conditions with racial diversity, would suggest non-White female applicants respond

racial diversity, gender diversity, or both racial and gender diversity?	having any racial diversity.	power to detect an OR = 1.72 (*d* = .30) main effect of racial diversity.	variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 51-60.	positively to seeing ingroup racial diversity reflected in organizational materials. 2) A main effect of gender diversity and a main effect of racial diversity, such that racial and gender diversity is associated with higher application rates, would suggest non-White female applicants respond positively to seeing both ingroup racial and ingroup gender diversity reflected in organizational materials. 3) A main effect of gender diversity, such that application rates are higher in conditions with gender diversity, would suggest non-White female applicants respond positively to seeing ingroup gender diversity reflected in organizational materials. 4) Null results showing neither main effect of gender nor racial diversity would suggest no evidence

				that our manipulations impacted application rates of non-White women.
Question 3: How do members of a stigmatized group (non-White women) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 6: Non-White women will apply more to organizations that are presented as having any level of either racial or gender diversity.	Data collection will continue until a minimum of 1524 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 2.07 (*d* = .40) interaction between racial and gender diversity.	Analysis 3: Logistic regression predicting the likelihood of applicant being a non-White woman from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 51-60.	1) See above for interpretations of main effects. 2) An interaction between gender and racial diversity would suggest that, for non-White females, the impact of the gender or racial diversity manipulation depends on the presence or absence of the other diversity manipulation. 3) Null results showing no gender by racial interaction would suggest that, for non-White females, there was no evidence that the presence of the racial diversity manipulation moderated the impact of the gender diversity manipulation (and vice versa).

Question 4: How do members of a non-stigmatized group (White men) respond to recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 7: White men will apply less to organizations that are presented as having greater racial diversity.	Data collection will continue until a minimum of 1036 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 0.58 (*d* = .30) main effect of racial diversity.	Analysis 4: Logistic regression predicting the likelihood of applicant being a White man from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 18-27.	1) A main effect of racial diversity, such that application rates are lower in conditions with racial diversity, would suggest White male applicants respond negatively to seeing outgroup racial diversity reflected in organizational materials. 2) A main effect of racial diversity, such that application rates are higher in conditions with racial diversity, would suggest White male applicants respond positively to seeing outgroup racial diversity reflected in organizational materials. 3) A main effect of gender diversity, such that application rates are lower in conditions with gender diversity, would suggest White male applicants respond negatively to seeing outgroup gender

				diversity reflected in organizational materials. 4) A main effect of gender diversity, such that application rates are higher in conditions with gender diversity, would suggest White male applicants respond positively to seeing outgroup racial diversity reflected in organizational materials. 5) Null results showing neither main effect of gender nor racial diversity would suggest no evidence that our manipulations impacted application rates of White men.
Question 4: How do members of a non-stigmatized group (White men) respond to recruitment materials displaying racial diversity, gender diversity, or	Hypothesis 8: White men will apply less to organizations that are presented as having any level of either racial or gender diversity.	Data collection will continue until a minimum of 1524 eligible applications are submitted, which is expected to provide greater than 95% power to detect an OR = 0.48 (*d* = .40) interaction between	Analysis 4: Logistic regression predicting the likelihood of applicant being a White man from gender diversity condition, racial diversity condition, interaction between conditions, and dummy-coded variable controlling for position applied for. See https://osf.io/8g65r/, lines 8-14 and 18-27.	1) See above for interpretations of main effects. 2) An interaction between gender and racial diversity would suggest that, for White males, the impact of the gender or racial diversity manipulation on application rates depends

both racial and gender diversity?		racial and gender diversity.		on the presence or absence of the other diversity manipulation. 3) Null results showing no gender by racial interaction would suggest that, for White males, there was no evidence that the presence of the racial diversity manipulation moderated the impact of the gender diversity manipulation (and vice versa) on application rates.
Question 5: How is the quality of applicants from a stigmatized group (White women) impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 9: The average quality of White female applicants will decrease when organizations are presented as having gender diversity.	Data collection will continue until there is an average of 42 eligible applicants per condition, which will provide 95% power for detecting a main effect of d = .57 for either racial or gender diversity.	Analysis 5: 2 (Racial Diversity) by 2 (Gender Diversity) between-subjects ANOVA on coded applicant quality. See https://osf.io/m2fd4/, lines 7-13 and 35-48.	1) A main effect of gender diversity, such that application quality decreases in conditions with gender diversity, would suggest that recruitment materials displaying gender diversity increase organizational appeal among less-qualified White female applicants. 2) A main effect of gender diversity, such that application quality increases in conditions with

			gender diversity, would suggest that recruitment materials displaying gender diversity increase organizational appeal among more-qualified White female applicants. 3) An interaction between racial and gender diversity would suggest that, for White females, the impact of the gender diversity manipulation on applicant quality depends on the presence or absence of racial diversity also present in organizational materials. 4) Null results for main effects of the racial and gender diversity manipulation would suggest no evidence that either manipulation impacted applicant quality of White females. A null result for the interaction would suggest that, for White females, there was no evidence that the impact of the racial diversity

				manipulation on applicant quality was moderated by the presence of the gender diversity manipulation (and vice versa).
Question 6: How is the quality of applicants from a stigmatized group (non-White men) impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 10: The average quality of non-White male applicants will decrease when organizations are presented as having greater racial diversity.	Data collection will continue until there is an average of 74 eligible applicants per condition, which will provide 95% power for detecting a main effect of *d* = .42 for either racial or gender diversity.	Analysis 6: 2 (Racial Diversity) by 2 (Gender Diversity) between-subjects ANOVA on coded applicant quality. See https://osf.io/m2fd4/, lines 7-13 and 52-65.	1) A main effect of racial diversity, such that application quality decreases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among less-qualified non-White male applicants. 2) A main effect of racial diversity, such that application quality increases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among more-qualified non-White male applicants. 3) An interaction between racial and gender diversity

		would suggest that, for non-White males, the impact of the racial diversity manipulation on applicant quality depends on the presence or absence of gender diversity also present in organizational materials. 4) Null results for main effects of the racial and gender diversity manipulation would suggest no evidence that either manipulation impacted applicant quality of non-White males. A null result for the interaction would suggest that, for non-White males, there was no evidence that the impact of the racial diversity manipulation on applicant quality was moderated by the presence of the gender diversity manipulation (and vice versa).

Question 7: How is the quality of applicants from a stigmatized group (non-White women) impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 11: The average quality of non-White female applicants will decrease when organizations are presented as having greater racial diversity.	Data collection will continue until there is an average of 37 eligible applicants per condition, which will provide 95% power for detecting a main effect of *d* = .60 for either racial or gender diversity	Analysis 7: 2 (Racial Diversity) by 2 (Gender Diversity) between-subjects ANOVA on coded applicant quality. See https://osf.io/m2fd4/, lines 7-13 and 69-82.	1) A main effect of racial diversity, such that application quality decreases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among less-qualified non-White female applicants. 2) A main effect of racial diversity, such that application quality increases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among more-qualified non-White female applicants. 3) A main effect of gender diversity, such that application quality decreases in conditions with gender diversity, would suggest that recruitment materials

			displaying gender diversity increase organizational appeal among less-qualified non-White female applicants. 4) A main effect of gender diversity, such that application quality increases in conditions with gender diversity, would suggest that recruitment materials displaying gender diversity increase organizational appeal among more-qualified non-White female applicants. 5) An interaction between racial and gender diversity would suggest that, for non-White females, the impact of the racial or gender diversity manipulation on applicant quality depends on the presence or absence of the other diversity manipulation also present in the materials. 6) Null results for main effects of the racial and

				gender diversity manipulation would suggest no evidence that either manipulation impacted applicant quality of non-White females. A null result for the interaction would suggest that, for non-White females, there was no evidence that the impact of the racial diversity manipulation on applicant quality was moderated by the presence of the gender diversity manipulation (and vice versa).
Question 8: How is the quality of applicants from a non-stigmatized group (White men) impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 12: The average quality of White male applicants will increase when organizations are presented as having greater racial or gender diversity.	Coders will rate applicants once there is at least an average of 75 eligible applicants per condition, which will provide 95% power for detecting a main effect of d = .42 for either racial or gender diversity.	Analysis 8: 2 (Racial Diversity) by 2 (Gender Diversity) between-subjects ANOVA on coded applicant quality. See https://osf.io/m2fd4/, lines 7-13 and 18-31.	1) A main effect of racial diversity, such that application quality decreases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among less-qualified White male applicants.

		2) A main effect of racial diversity, such that application quality increases in conditions with racial diversity, would suggest that recruitment materials displaying racial diversity increase organizational appeal among more-qualified White male applicants. 3) A main effect of gender diversity, such that application quality decreases in conditions with gender diversity, would suggest that recruitment materials displaying gender diversity increase organizational appeal among less-qualified White male applicants. 4) A main effect of gender diversity, such that application quality increases in conditions with gender diversity, would suggest that recruitment materials displaying gender diversity increase

			organizational appeal among more-qualified White male applicants. 5) An interaction between racial and gender diversity would suggest that, for White males, the impact of the racial or gender diversity manipulation on applicant quality depends on the presence or absence of the other diversity manipulation also present in the materials. 6) Null results for main effects of the racial and gender diversity manipulation would suggest no evidence that either manipulation impacted applicant quality of White males. A null result for the interaction would suggest that, for White males, there was no evidence that the impact of the racial diversity manipulation on applicant quality was moderated by the presence of the gender

				diversity manipulation (and vice versa).
Question 9: How is the overall quantity of applicants to an organization impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 13: Overall rate of application submission from website visitors will increase for organizations displaying racial or gender diversity.	Data collection will continue until there are 1036 eligible applications submitted. Assuming 3% of website visitors submit applications, this sample size provides 95% power to detect OR = 1.42 (*d* = .19) for a main effect of racial diversity, gender diversity, or an interaction.	Analysis 9: Logistic regression predicting the likelihood of website visitor submitting an application from gender diversity condition, racial diversity condition, and interaction between conditions. See https://osf.io/2dyx9/, lines 8-14 and 24-28.	1) A main effect of racial diversity, such that applications increase in conditions with racial diversity, would suggest that website visitors were more interested in applying to organizations displaying racial diversity in recruitment materials. 2) A main effect of racial diversity, such that applications decrease in conditions with racial diversity, would suggest that website visitors were less interested in applying to organizations displaying racial diversity in recruitment materials. 3) A main effect of gender diversity, such that applications increase in conditions with gender diversity, would suggest that website visitors were more interested in applying

			to organizations displaying gender diversity in recruitment materials. 4) A main effect of gender diversity, such that applications decrease in conditions with gender diversity, would suggest that website visitors were less interested in applying to organizations displaying gender diversity in recruitment materials. 5) An interaction between racial and gender diversity would suggest that the impact of the racial or gender diversity manipulation on the number of applications submitted depends on the presence or absence of the other diversity manipulation also present in the materials. 6) Null results for main effects of the racial and gender diversity manipulation would

				suggest no evidence that either manipulation impacted number of applications submitted. A null result for the interaction would suggest no evidence that the impact of the racial diversity manipulation on the number of applications was moderated by the presence of the gender diversity manipulation (and vice versa).
Question 10: How is the overall quantity of application started impacted by recruitment materials displaying racial diversity, gender diversity, or both racial and gender diversity?	Hypothesis 14: Overall rate of applications started from website visitors will increase for organizations displaying racial diversity or gender diversity.	Data collection will continue until there are 1036 eligible applications submitted. Assuming 4% of website visitors start an application, this sample size provides 95% power to detect OR = 1.31 (*d* = .15) for a main effect of racial diversity, gender diversity, or an interaction.	Analysis 10: Logistic regression predicting the likelihood of website visitor starting an application from gender diversity condition, racial diversity condition, and interaction between conditions. See https://osf.io/2dyx9/, lines 8-14 and 18-22.	1) A main effect of racial diversity, such that applications increase in conditions with racial diversity, would suggest that website visitors were more interested in starting applications to organizations displaying racial diversity in recruitment materials. 2) A main effect of racial diversity, such that applications decrease in conditions with racial

			diversity, would suggest that website visitors were less interested in starting applications to organizations displaying racial diversity in recruitment materials. 3) A main effect of gender diversity, such that applications increase in conditions with gender diversity, would suggest that website visitors were more interested in starting applications to organizations displaying gender diversity in recruitment materials. 4) A main effect of gender diversity, such that applications decrease in conditions with gender diversity, would suggest that website visitors were less interested in starting applications to organizations displaying gender diversity in recruitment materials.

		5) An interaction between racial and gender diversity would suggest that the impact of the racial or gender diversity manipulation on the number of applications started depends on the presence or absence of the other diversity manipulation also present in the materials. 6) Null results for main effects of the racial and gender diversity manipulation would suggest no evidence that either manipulation impacted number of applications started. A null result for the interaction would suggest no evidence that the impact of the racial diversity manipulation was moderated by the presence of the gender diversity manipulation (and vice versa).

***** - The threshold for determining statistical significance will be set at $\alpha = 0.05$.

Table 2. Resources available in online supplement

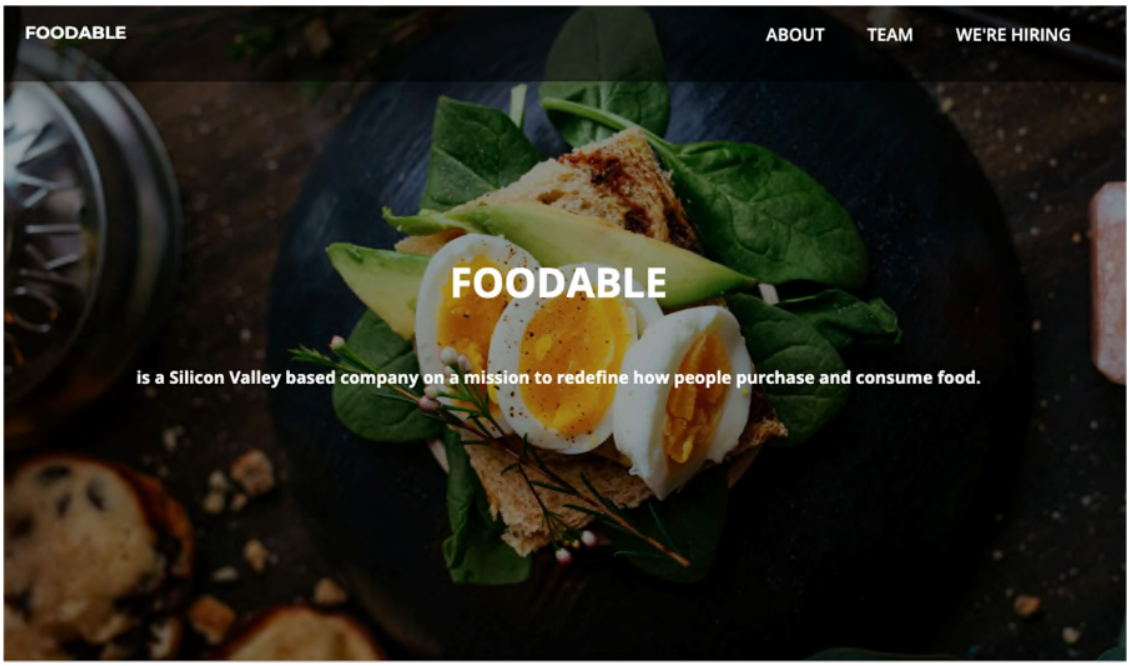
File Name	Description	Direct Link
Demographic Distributions, Condition Assignment and Maintenance		
DemographicDistributionInfo	Summary of report for demographics of tech workers used to calculate anticipated sample distributions	https://osf.io/jtvnf/
TechSectorDiversityDemographics	Source data for calculating anticipated sample distributions based on demographics	https://osf.io/xtucw/
ConditionAssignmentMaintenance	Information on how participants are assigned to conditions across devices and IP addresses	https://osf.io/39fwt/
Power Analyses		
PowerAnalyses.AdditionalInfo	Background information for all subgroups used to calculate power analyses	https://osf.io/8asqp/
Power.NonWhiteMen.Option1	Power analysis for non-White men anticipating only a main effect of racial diversity	https://osf.io/fdk2g/
Power.NonWhiteMen.Option2	Power analysis for non-White men anticipating an interaction between racial and gender diversity	https://osf.io/fb3cv/
Power.NonWhiteWomen.Option1	Power analysis for non-White women anticipating only a main effect of racial diversity	https://osf.io/pqd65/
Power.NonWhiteWomen.Option2	Power analysis for non-White women anticipating an interaction between racial and gender diversity	https://osf.io/3hqmu/
Power.WhiteMen.Option1	Power analysis for White men anticipating only a main effect of racial diversity	https://osf.io/a5ymp/
Power.WhiteMen.Option2	Power analysis for White men anticipating an interaction between racial and gender diversity.	https://osf.io/uy5m8/
Power.WhiteWomen.Option1	Power analysis for White women anticipating only a main effect of gender diversity	https://osf.io/wzfyj/
Power.WhiteWomen.Option2	Power analysis for White women anticipating an interaction between racial and gender diversity	https://osf.io/zy6gh/

Sample Data and Analysis Scripts		
Analysis.ApplicantDiversity.SampleData	Sample data for analysis concerning applicant diversity	https://osf.io/havq2/
Analysis.ApplicantDiversity.Syntax	Syntax for analysis concerning applicant diversity	https://osf.io/4hrjk/
Analysis.ApplicantQuality.SampleData	Sample data for analysis concerning applicant quality	https://osf.io/8qsrg/
Analysis.ApplicantQuality.Syntax	Syntax for analysis concerning applicant quality	https://osf.io/d5r2u/
Analysis.TotalApplications.SampleData	Sample data for analysis concerning total applications started or completed	https://osf.io/kjfah/
Analysis.TotalApplications.Syntax	Syntax for analysis concerning total applications started or completed	https://osf.io/9v4su/
Analysis.Equivalence Tests	Syntax for equivalence tests to be run on any null results relevant for Hypotheses 1-14 specified in Table 1.	https://osf.io/qtdxk/

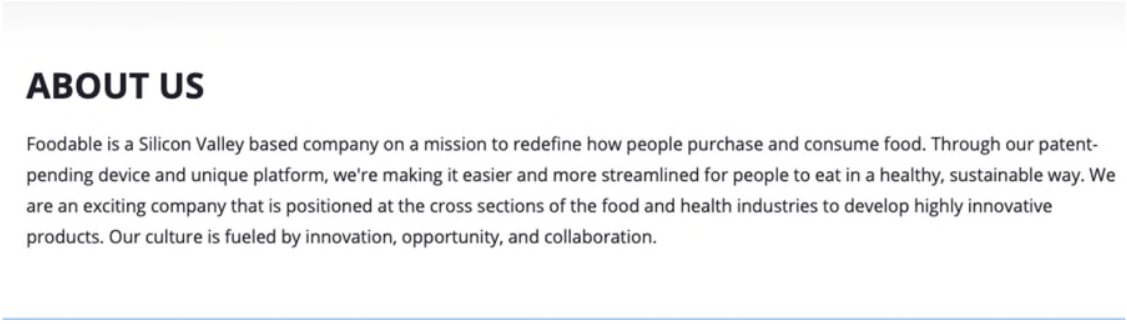
Appendix A

Website content for “Foodable” (https://foodabletech.com/).

Welcome Screen



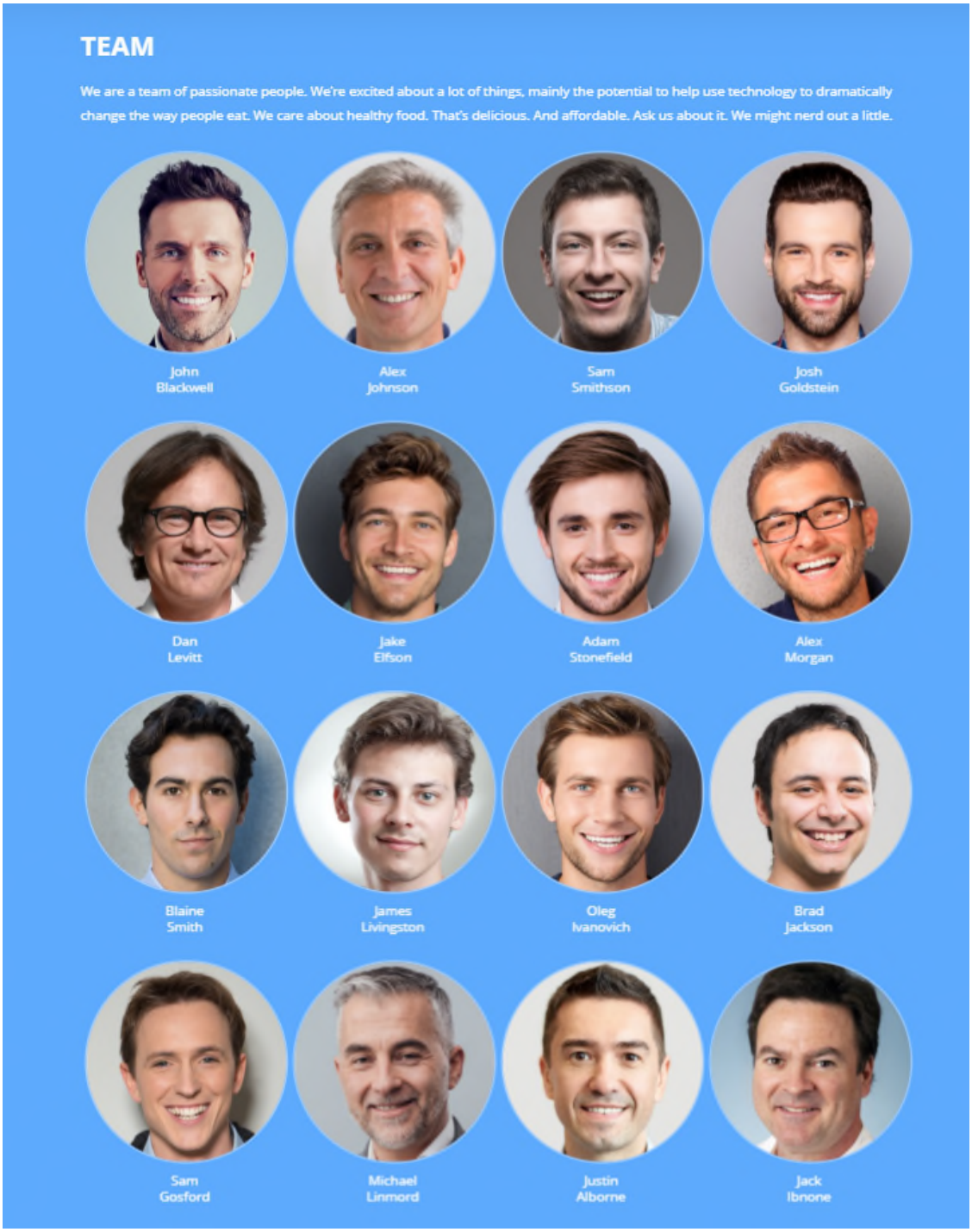
‘About Us’ Section



ABOUT US

Foodable is a Silicon Valley based company on a mission to redefine how people purchase and consume food. Through our patent-pending device and unique platform, we're making it easier and more streamlined for people to eat in a healthy, sustainable way. We are an exciting company that is positioned at the cross sections of the food and health industries to develop highly innovative products. Our culture is fueled by innovation, opportunity, and collaboration.

Sample ‘Team’ Section

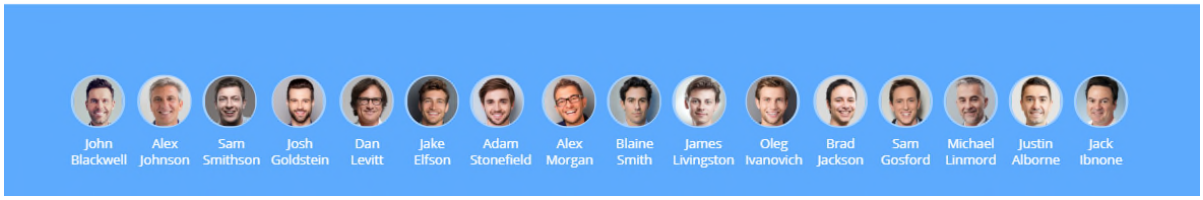


We’re Hiring Section

WE'RE HIRING

These are the current positions open. All positions are based at our location in Silicon Valley; we will cover relocation costs for the right candidates.

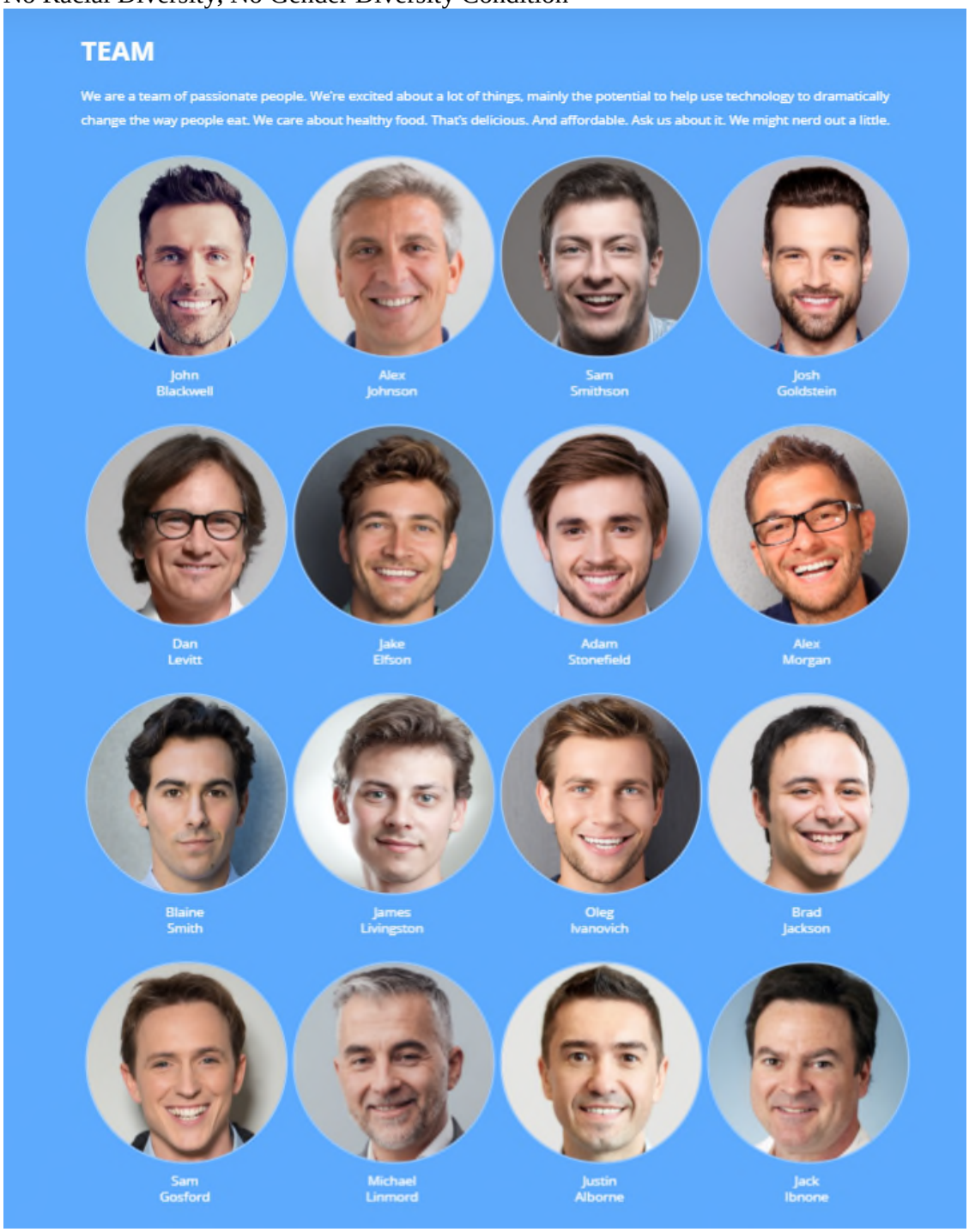
- Senior Product Manager
- Marketing Communications Manager
- Embedded Systems Architect
- Full Stack Engineer



Appendix B

Experimental conditions.

No Racial Diversity, No Gender Diversity Condition



No Racial Diversity, Gender Diversity Condition

No Gender Diversity, Racial Diversity Condition

Gender Diversity, Racial Diversity Condition

Appendix C

Job description information for available positions.

Senior Product Manager

Senior Product Manager

ROLE

The Sr. Product Marketing Manager will develop an intimate understanding of the market and which use-cases and segments to prioritize. You will be responsible for creating a product roadmap that addresses these segments and identifying the right feature set for each release.

RESPONSIBILITIES

- Identify and define key use cases and market segments to pursue for new products and to grow our current offerings in.
- Develop the product roadmap and lead the planning, definition and prioritization of features for future product releases by conducting market research.
- Help define the go-to-market strategy for each product release.
- Identify and evaluate potential partnerships that support the go-to-market strategy.
- Develop the core value proposition for the product and it's respective positioning and messaging.
- Be an expert with respect to competitive offerings in order to deliver products recognized as thought leading in the industry.
- Define business models for product offerings and set the corresponding pricing strategy.
- Work with finance to produce revenue forecasts
- Define and analyze metrics that inform the success of your product.

QUALIFICATIONS

- Entrepreneurial spirit with a strong desire to make the world a better place.
- A minimum of a Bachelor's degree in Marketing, Business or a related field is required. An MBA is preferred but not required.
- A minimum of 5 years of product management experience is required.
- Demonstrated success in defining and launching new products that include a hardware component.
- Proven ability to influence cross-functional teams without formal authority.
- Experience managing and prioritizing multiple projects is required.
- Excellent communication skills (written and oral) are required.

BENEFITS

We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

APPLY HERE

Marketing Communications Manager

Marketing Communications Manager

ROLE

The Marketing Communications Manager is a communication specialist is responsible for executing a wide range of informational materials for internal and external audiences. The Marketing Communications Manager will manage the development and execution of go-to market strategies and campaigns that drive awareness, engagement and adoption of our solutions.

RESPONSIBILITIES

- Develop and deliver impactful, strategic marketing communications programs and campaigns in support of business goals, such as brand differentiation, market awareness, demand generation and sales enablement with prospects as well as existing customers.
- Work cross-functionally with marketing program managers, product marketing and creative and web services teams to facilitate successful execution of new product launches.
- Partner with the Business Unit to develop multi-channel and multi-media marketing content in support of marketing objectives (including but not limited to press releases, website content, product collateral including datasheets, brochures, white papers and success stories, webinars, videos, contributed editorial articles and advertisements).
- Use digital marketing technology and techniques, including marketing automation and customer engagement platforms, search engine marketing (SEM), search engine optimization (SEO) and social media tools to promote and market products and services.
- Develop and implement methods of measuring marketing effectiveness and ROI.
- Provide leadership in managing events of various sizes, working cross-functionally with marketing program management, product marketing and an events management team.

QUALIFICATIONS

- Entrepreneurial spirit with a strong desire to make the world a better place.
- A minimum of a Bachelor's degree in Marketing, Business or a related field is required.
- A minimum of 3 years of marketing experience is required, including experience in digital marketing.
- Experience managing and prioritizing multiple projects is required.
- Excellent communication skills (written and oral) are required.

BENEFITS

We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

APPLY HERE

Embedded Systems Architect

Embedded Systems Architect

ROLE

As the Embedded Systems Architect, you are the key person in driving various aspects of our hardware designs that range from small durable form-factors such as mobile sensors to medical grade equipment. You will collaborate with many internal cross-functional teams from compliance and certification to service software engineering teams to test and QA. In order for you to succeed, strong communication skills and a strong customer oriented passion are a must. Our team values open collaboration and positive work environments.

RESPONSIBILITIES

- Design hardware for mobile devices, commercial food grade products and medical devices
- Contribute to cross functional designs including enclosures, pcb layout and efficient RTOS software
- Drive decisions for quality designs that balance, manufacturability, technology longevity and cost.
- Actively participate in design reviews.
- Support products through their lifecycle.

QUALIFICATIONS

- Degree in Electrical Engineering, Computer Science or related field experience with 5+ years in consumer facing embedded systems required. Masters Degree or PhD preferred.
- Experience with sensor design and radio technologies is a must.
- Experience with Medical and low-cost small form factor designs are a strong plus.
- Solid communication skills coupled with technical document writing is a must.
- Solid track record of designing and building low-cost manufacturable devices is required.

BENEFITS

We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

APPLY HERE

Full Stack Engineer

Full Stack Engineer

ROLE

As the Senior Full Stack Engineer, you are empowered to build our services from backend databases all the way to mobile and web-clients. Since our team is small and growing, picking the right technology choices for our service oriented architecture that can scale to a variety of clients will be your primary goal. Fast, Scalable and Secure are the key.

RESPONSIBILITIES

- Build robust fault tolerant, scalable, secure and efficient services end to end.
- Willingness to dive into the client side code for minor and major features.
- Collaborate closely with cross functional teams from hardware, Site and Reliability and frontend engineers.
- Comfortable working in Unix environments.
- Participate in design reviews.
- Drive design decisions for technology choices across the stack.
- Integrate and develop performance analytics. Design products that alert us to problems before the customer knows.

QUALIFICATIONS

- Degree in Computer Science, Computer Engineering or related field experience with 5+ years in full stack development is required. Masters Degree or PhD preferred.
- Expertise in Javascript, HTML and MVC technologies Rails, Django, Twisted, etc)
- Some experience in Java, Go or Python for backend services required.
- Strong understanding of service oriented architecture and how design decisions impact various clients is a must.
- Proven ability to learn new technologies, whether backend, frontend or databases to find the right solution for the team.
- Experience with performance analysis is a plus.
- Developing HIPAA compliant systems is a strong plus.

BENEFITS

We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

APPLY HERE

Web Developer

This role is particularly well-suited for someone who values creating high quality web pages and user experiences, as well as someone who has a strong sense of ownership of their product.

RESPONSIBILITIES

- Build out full-stack website features to help drive growth and engagement
- Maintain and grow our front facing web application
- Help grow our business through quick experimentation and feature development
- Data analytics and instrumentation using Google Analytics
- Publish content to our website using CMS

QUALIFICATIONS

- A demonstrated ability to produce high-quality code
- Computer Science background and understanding of algorithms and data structures.
- Experience with modern software development including pull requests, continuous integration/deployment, unit testing, and agile methodologies.
- Experience with cloud technologies.
- Experience writing SQL and tuning relational database queries.
- Proficiency in creating integrations with external APIs.
- Experience in front-end technology and framework such as HTML, CSS, JavaScript, BackboneJS, ReactJS or similar.
- Familiar with development concepts such as Dependency Injection, Mocking/Test Doubles, Object Relational Mappers.
- Appreciation for testing and the ability to write specs.
- Experience working and collaborating with designers.
- Experience with data analytics tools like Google Analytics, or similar, is preferred.
- Strong prioritization skills

BENEFITS

We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

Account Manager

ROLE
The Account Manager will be in charge of maintaining current client relationships, and develop new business from existing accounts.

RESPONSIBILITIES
- Deliver top-notch customer service by providing clients with prompt responses and helping meet their needs.
- Identify new business opportunities within existing customer base.
- Provide high level product training for customers.
- Manage growing book of customers.
- Understand the functions of our product and mission and be able to articulate our key features to potential clients.
- Stay up to date on relevant products and technology.
- Build and maintain strong client relationships.
- Prepare and submit monthly account forecasts.

QUALIFICATIONS
- Results-driven focus to satisfy existing clients.
- A minimum of a Bachelor's degree is required.
- Familiarity selling to large corporations across different industries.
- Knowledge of accounting and financial concepts is preferred.
- Proven history of success in technology or a related field.
- Working knowledge of Microsoft Office.

BENEFITS
We offer a competitive benefits package including vacation, ho

Business Analyst

ROLE
The Business Analyst will play a vital role in coordinating between the business team and other teams throughout Foodable.

RESPONSIBILITIES
- Analyze and synthesize information provided by stakeholders.
- Lead ongoing reviews of business processes and developing optimization strategies.
- Conduct interviews and independent research to design solutions for business problems.
- Stay up to date on latest advancements to modernize Foodable systems.
- Document, track, and resolve issues related to data quality.
- Remain available daily for questions from other project members.
- Work closely with clients, technicians and managerial staff.
- Lead or facilitate business requirement workshops.
- Effectively communicate insights to team members and management.

QUALIFICATIONS
- Detail-oriented, analytical mindset to find issues and identify solutions.
- A Bachelor’s degree in business or a related field.
- Ability to influence without direct authority.
- Adaptable to changes in a fast-paced environment.
-Working knowledge of Microsoft Office
- Exceptional analytical and conceptual thinking skills
- Excellent documentation skills

BENEFITS
We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

Copywriter

ROLE
Your writing will shape user interfaces, enhance UX, define product value propositions, and clarify in-app messaging. It will also support marketing initiatives.

RESPONSIBILITIES
- Take complicated product challenges and translate them into compelling, clear brand stories that are understandable and interesting.
- Support writing needs across multiple projects, including upcoming campaign work, product messaging, social and digital copy and pitch decks.
- Collaborate with the visual design team, the digital/social team, and the product marketing teams to express our voice in a variety of projects.
- Be comfortable working as the lead writer, with direction from Product Managers, to translate strategy into compelling copy in both short and long form.
- Combine a passion for copywriting and copyediting with a drive to create communication that makes readers take action, using clear and memorable consumer-facing language.

QUALIFICATIONS
- Proven advanced writing skills
- Strong editing and proofreading skills.
- The ability to work quickly and meet tight deadlines on a multitude of concurrent tasks.
- The ability to work well both independently and in a team. Experience managing and prioritizing multiple projects is required.
- The ability (and desire) to ideate on things like campaign themes, ad concepts, and product names.

BENEFITS
We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

Recruiter

ROLE
The Recruiter role will be in charge of coordinating the hiring aspects of the company. You will be responsible for using various resources and platforms to locate the right candidates for our growing team.

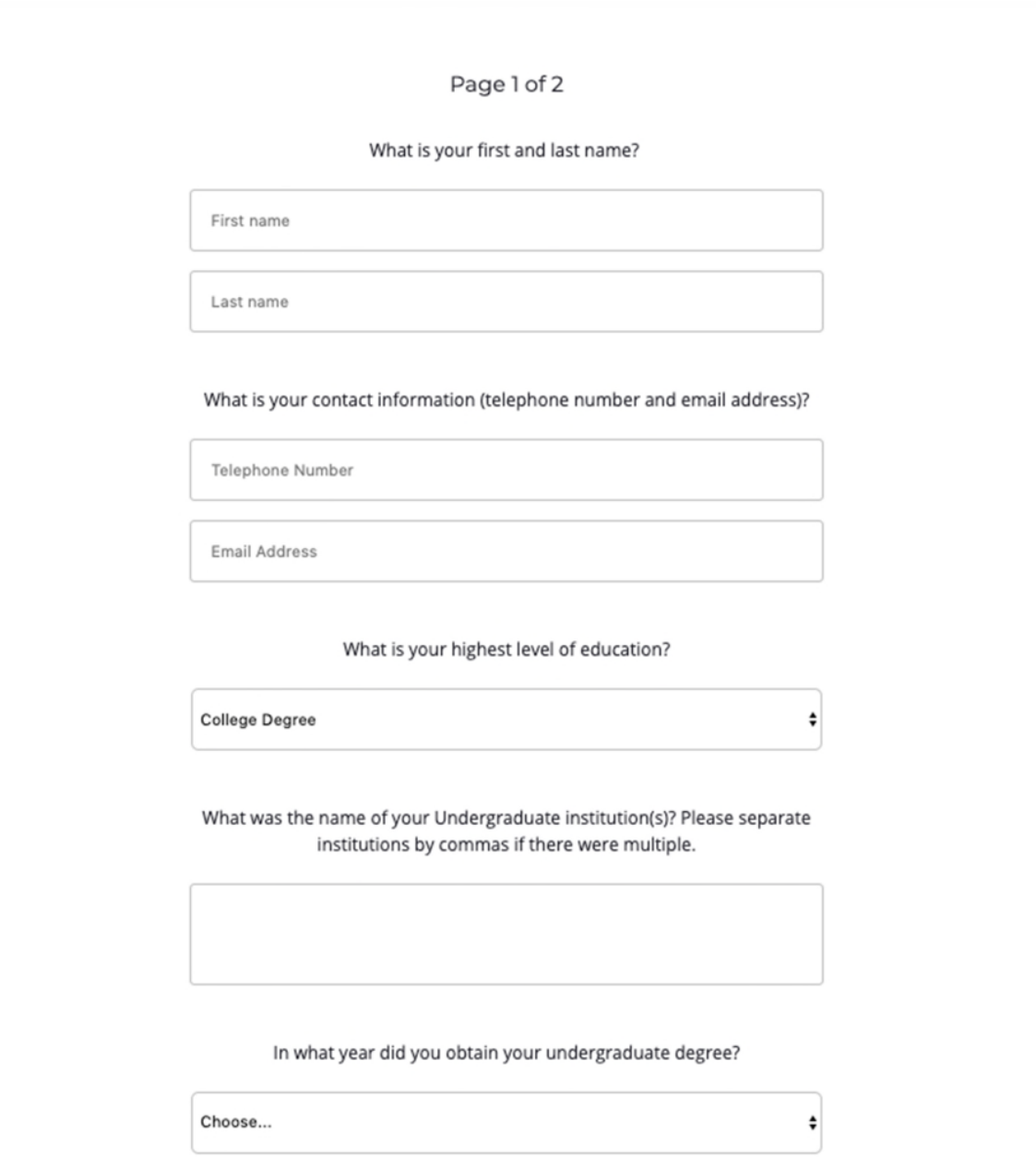
RESPONSIBILITIES
- Manage full-cycle employee recruitment across multiple teams.
- Generate a diverse candidate population using methods like including job boards, membership organizations, and personal networking.
- Create efficient and effective recruitment strategies.
- Conduct phone interviews with applicants to ensure candidates meet minimum qualifications.
- Screen candidate applications to ensure qualifications.
- Research new tools for effective recruitment and best practices.
- Provide information to potential applicants about the Foodable mission and team.
- Communicate status and progress on filling open positions.
- Ensure a smooth application process for candidates.

QUALIFICATIONS
- Motivated attitudes to go the extra mile to find the right candidate.
- Some prior experience in recruitment is preferred.
- Demonstrated success in facilitating with the screening and hiring of potential candidates.
- Proven ability to handle sensitive and confidential information.
- Experience managing multiple ongoing projects is required.
- Excellent communication skills (written and oral) are required.

BENEFITS
We offer a competitive benefits package including vacation, holidays, 401(k), medical, and dental. Food and snacks are provided on our premises.

Appendix D

Sample application portal (for stack engineer position).



Page 1 of 2

What is your first and last name?

First name

Last name

What is your contact information (telephone number and email address)?

Telephone Number

Email Address

What is your highest level of education?

College Degree

What was the name of your Undergraduate institution(s)? Please separate institutions by commas if there were multiple.

In what year did you obtain your undergraduate degree?

Choose...

In total, how many semesters did you study as an undergrad?

During your undergraduate semesters, how many total semesters did you have a part-time job in addition to being a student?

What was your undergraduate GPA (overall)?

How many STEM (Science, Technology, Engineering and Math) classes did you take as an undergraduate student?

What was your GPA for your undergraduate STEM (Science, Technology, Engineering and Math) classes?

How many non-STEM (Science, Technology, Engineering and Math) classes did you take as an undergraduate?

What was your GPA for the non-STEM (Science, Technology, Engineering, and Math) classes you took as an undergraduate?

Please list all awards or honors you received as an undergraduate student. Please separate achievements using commas.

Overall, how would you rate your performance as a student during your undergraduate years?

Awful (1) Excellent (10)

In total, how many years of work experience do you have?

NEXT

Please answer the following questions about the jobs or internships you've held, starting with the most recent.

Role/Position/Job or Internship Title

Company

In total, how long did you work in this role?

Years | Months

years | months

Please write a brief description of the responsibilities/achievements. Copying and pasting bullet points from your resume is fine.

Overall, how would your rate your performance at this company?

Awful (1) — Excellent (10)

Would you like to add information about another job or internship you have held?

ADD JOB OR INTERNSHIP

Please rate your skill level in each domain where '0' = No skill and '10' = Expert skill.

In Javascript

No Skill (1) Expert Skill (10)

In Jango

No Skill (1) Expert Skill (10)

In Rails

No Skill (1) Expert Skill (10)

We've prepared a list of articles about the food industry's market dynamics, trends, and challenges. Would you be interested in reading these articles, if they were made available?

- Yes, I am interested in receiving articles about the food industry market
- No, I am not interested in receiving articles about the food industry market

Our founder was recently invited to speak on a webinar about successful approaches to launching innovative tech. Would you be interested in watching this webinar, if it was made available?

- Yes, I am interested in watching a webinar about launching innovative tech
- No, I am not interested in watching a webinar about launching innovative tech

For planning purposes, please let us know at which times (EST) you would be willing to interview for the position to which you have applied.

- [ ] 6:00am - 7:00am
- [ ] 7:00am - 8:00am
- [ ] 8:00am - 9:00am
- [ ] 9:00am - 10:00am
- [ ] 10:00am - 11:00am
- [ ] 11:00am - 12:00am
- [ ] 12:00am - 13:00am
- [ ] 13:00am - 14:00am
- [ ] 14:00am - 15:00am
- [ ] 15:00am - 16:00am
- [ ] 16:00am - 17:00am
- [ ] 17:00am - 18:00am
- [ ] 18:00am - 19:00am
- [ ] 19:00am - 20:00am
- [ ] 20:00am - 21:00am
- [ ] I am not willing to be interviewed for this position

Note: We will not use this information when evaluating job applicants. We are collecting this information for statistical purposes only:

What is your gender?

- ( ) Male
- ( ) Female
- ( ) Other

What is your race/ethnicity? Please check all that apply:

- [ ] Black or African American
- [ ] Asian or Pacific Islander
- [ ] White
- [ ] Latino/a/x
- [ ] Native American or American Indian
- [ ] Middle Eastern or North African Descent
- [ ] Other

SUBMIT

[Planned addition to the application portal]

How did you hear about this opportunity?

- Online Advertisement
- Online Search
- Referral

Appendix E

Rubric used by human resource coders to evaluate applicants.

Candidate:	DATE
Evaluator:	

RUBRIC	SCORE
Extremely Acceptable; Highest skill set; Extremely strong	5
Very Acceptable; High skill set; Very strong	4
Acceptable; Moderate strong skill set; Moderately strong	3
Slightly Acceptable; Guidelines somewhat met; Somewhat strong	2
Not Acceptable; Guidelines not met; Not at all strong	1

SCORING SCALE	TOTAL
EXTREMELY ACCEPTABLE	5
VERY ACCEPTABLE	4
ACCEPTABLE	3
SLIGHTLY ACCEPTABLE	2
NOT ACCEPTABLE	1

CRITERIA	5	4	3	2	1
Interpersonal Skills: Does the candidate possess the required technical skills to perform the job OR have they demonstrated the ability to perform these skills?					
Education/Knowledge: Does the candidate demonstrate strong work experience or internship?					
Knowledge and Skills in Research: Does the candidate demonstrate clear understanding / with interested in webinars and receiving articles about the industry?					
Leadership and Collegiality: Has experience in effectively working with others on various types of projects.					

Supplementary Materials

Details on Randomization and Maintenance of Condition Assignment

When participants open the link to the website, they will be randomized to see one of the four experimental conditions. Specifically, participants will be randomized to view the version of the website with the least number of assigned visitors. Two technical factors will be utilized to maintain condition assignment: HTTP cookies and IP addresses. HTTP cookies are data that websites send to the user's browser that allows the user's browser to quickly reload website information for later revisits. Importantly, browser-stored HTTP cookies will allow the website to identify the participant's assigned website condition across multiple visits. IP addresses are numerical network identifiers used to differentiate internet users across various internet access points. Critically, IP addresses for devices on the same internet network are identical; that is, users accessing the website from their home WiFi will have the same IP address whether they are accessing the website via laptop or mobile phone. The technical components of the randomization and assignment procedure are detailed further below.

After participants open the website link, they will be randomized to their website condition. In the background, participants' browsers are sent HTTP cookies by accessing the website. Next, the website links the information about condition assignment with the participants' cookies and IP address. This two-factor authentication ensures that every new visitor that has either a) the same cookies and/or b) the same IP address will see the version of the website they saw at first visit. If the participant disables HTTP cookies or clears their cookies between visits, they will view the website condition assigned to their IP address.[1] Ultimately, maintaining assignment using IP addresses helps prevent participants from detecting condition within networks (i.e., across devices at home, at the office, at the library), while also maintaining assignment with HTTP cookies helps prevent participants from detecting condition between networks when the same device-browser combination is used. For concrete examples of instances for which assignment would or would not be maintained, see Supplementary Table 1 below.

[1] Note: Website visitors do not have to accept browser cookies. Some browsers have private browsing modes (i.e., Incognito mode on Google Chrome) or settings that prevent websites from storing cookies on participants' computers. Of course, users may also elect to clear their cookies and browsing history at any point. However, our discussions with the developers indicated that internet users do not frequently reject or clear all cookies between browsing sessions.

Supplementary Table 1. Condition Assignment Examples

VISIT at Time 1				VISIT at Time 2					
Device 1	**Browser 1**	**Cookies**	**Network**	**Device 2**	**Browser 2**	**Cookies**	**Network**	**Assignment**	**Reason**
Phone	Chrome	Accepted	WiFi 1	Phone	Chrome	Accepted	WiFi 1	Maintained	Same IP Address, Same Cookies
Phone	Chrome	Accepted	WiFi 1	Phone	Chrome	Accepted	Cellular Network (LTE)	Maintained	Same Cookies
Phone	Chrome	Accepted	WiFi 1	Laptop	Chrome	Accepted	WiFi 1	Maintained	Same IP Address
Phone	Chrome	Accepted	WiFi 1	Laptop	Firefox	Accepted	WiFi 1	Maintained	Same IP Address
Phone	Chrome	Accepted	WiFi 1	Laptop	Firefox	Not Accepted or Later Cleared	WiFi 1	Maintained	Same IP Address
Phone	Chrome (Incognito)	Not Accepted or Later Cleared	WiFi 1	Laptop	Firefox	Not Accepted or Later Cleared	WiFi 1	Maintained	Same IP Address
Phone	Chrome (Incognito)	Not Accepted or Later Cleared	WiFi 1	Laptop	Firefox	Not Accepted or Later Cleared	WiFi 1	Maintained	Same IP Address
Phone	Chrome	Accepted	WiFi 1	Laptop	Firefox	Not Accepted or Later Cleared	Cellular Network (LTE)	Not Maintained	Different IP Addresses & Different Cookies
Phone	Chrome	Accepted	WiFi 1	Phone	Firefox	Accepted	Cellular Network (LTE)	Not Maintained	Different IP Addresses & Different Cookies
Phone	Chrome (Incognito)	Not Accepted or Later Cleared	WiFi 1	Phone	Chrome	Accepted	Cellular Network (LTE)	Not Maintained	Different IP Addresses & No Cookies at T1

Note: This table does not exhaust all possible assignment scenarios. However, it is meant to illustrate that condition assignment will be maintained as long as: a) cookies are kept and remain constant across visits or b) IP address is constant across site visits.

Additional Notes on the Demographics Statistics Used for Power Calculations

This project utilized demographic data from 177 high tech companies collected by "Reveal from The Center for Investigative Reporting and The Center for Employment Equity." [https://www.revealnews.org/blog/how-we-created-a-baseline-for-silicon-valleys-diversity-problem/]

The investigation referenced above used 2016 EEO-1 forms (federally mandated diversity reports) to find the following demographic breakdown of all workers across 177 tech corporations:

%White Women - ~16.2%
%White Men - ~41.26%
%non-White Men - ~28.53%
%non-White women - ~14.02%

To calculate the statistics for non-White men and non-White women, two simple calculations must be conducted on the publicly available dataset (See original dataset here: https://github.com/cirlabs/Silicon-Valley-Diversity-Data).

In the excel file labeled "Tech_sector_diversity_demographics_2016.csv,"

1. Subtract the percentage of White-female workers (Cell: E3) from the percentage of all female workers (Cell: E11) to observe the percentage of non-white female workers in technology.

2. Similarly, subtract the percentage of White-male workers (Cell: E2) from the percentage of all male workers (Cell: E10) to observe the percentage of non-white male workers in technology.

Online Supplement

Additional Information: Power analyses for each subgroup

Power Analyses for White Men (Based on 50,000 simulations)
Possibility 1: Only Racial Diversity is Threatening: In this scenario, we would expect lower rate of responses in any condition that includes racial diversity. The key analysis would involve detecting a reliable main effect of racial diversity. The analyses below would be the same if the effect were about gender, and the key analysis would involve detecting a reliable main effect of gender diversity.

If OR = 0.580 (Cohen's d = .30), expected distributions would be 41.26% in conditions without racial diversity, and 28.95% in conditions with racial diversity. A sample size of 700 would achieve 82.7% power to detect a main effect. A sample size of 944 would achieve 96.1% power to detect a main effect.

Possibility 2: All Diversity is Threatening: In this scenario, we would expect equally lower rates of responses in all conditions but the No Race Diversity / No Gender Diversity condition. The key analysis would involve detecting a reliable interaction between our racial and gender diversity conditions.

If OR = 0.484 (Cohen's d = .40), expected distributions would be 41.26% in the No Race, No Gender diversity condition, but 25.37% in all other conditions. A sample size of 1252 would achieve 84.7% power to detect an interaction effect. A sample size of 1512 would achieve 96.1% power to detect an interaction effect.

Power Analyses for Non-White Men (Based on 50,000 simulations)
Possibility 1: Greater responding to organizations with employees that share a stigmatized identity: In this scenario, we would expect higher rates of responses in any condition that includes racial diversity. The key analysis would involve detecting a reliable main effect of racial diversity.

If OR = 1.723 (Cohen's d = .30), expected distributions would be 28.53% in conditions without racial diversity, and 40.75% in conditions with racial diversity. A sample size of 512 would achieve 81.6% power to detect a main effect. A sample size of 720 would achieve 95.6% power to detect a main effect.

Possibility 2: Greater responding to organizations with employees that have any stigmatized identity: In this scenario, we would expect higher rates of responses in all conditions but the No Race Diversity / No Gender Diversity condition. The key analysis would involve detecting a reliable interaction between our racial and gender diversity conditions.

If OR = 2.066 (Cohen's d = .40), expected distributions would be 28.53% in the No Race, No Gender diversity condition, but 45.20% in all other conditions. A sample size of 1080 would achieve 80.4% power to detect an interaction effect. A sample size of 1500 would achieve 97.2% power to detect an interaction effect.

Power Analyses for White Women (Based on 50,000 simulations)
Possibility 1: Greater responding to organizations with employees that share a stigmatized identity: In this scenario, we would expect higher rates of responses in any condition that includes gender diversity. The key analysis would involve detecting a reliable main effect of gender diversity.

If OR = 1.723 (Cohen's d = .30), expected distributions would be 16.2% in conditions without gender diversity, and 24.99% in conditions with gender diversity. A sample size of 688 would achieve 81.0% power to detect a main effect. A sample size of 908 would achieve 97.5% power to detect a main effect.

Possibility 2: Greater responding to organizations with employees that have any stigmatized identity: In this scenario, we would expect higher rates of responses in all conditions but the No Race Diversity / No Gender Diversity condition. The key analysis would involve detecting a reliable interaction between our racial and gender diversity conditions.

If OR = 2.066 (Cohen's d = .40), expected distributions would be 16.2% in the No Race, No Gender diversity condition, but 28.54% in all other conditions. A sample size of 1208 would achieve 82.7% power to detect an interaction effect. A sample size of 1524 would achieve 96.2% power to detect an interaction effect.

<u>Power Analyses for Non-White Women (Based on 50,000 simulations)</u>
Possibility 1: Greater responding to organizations with employees that share the same racial identity: In this scenario, we would expect higher rates of responses in any condition that includes racial diversity. The key analysis would involve detecting a reliable main effect of racial diversity. We draw this prediction from papers finding that women from racial minority groups responded more positively to organizations that highlighted a member who shared their racial compared to their gender background (e.g., Johnson, Pietri, Fullilove & Mowrer, 2019).

If OR = 1.723 (Cohen's d = .30), expected distributions would be 14.02% in conditions without racial diversity, and 21.93% in conditions with racial diversity. A sample size of 692 would achieve 83.2% power to detect a main effect. A sample size of 1036 would achieve 95.9% power to detect a main effect.

Possibility 2: Greater responding to organizations with employees that have any stigmatized identity: In this scenario, we would expect higher rates of responses in all conditions but the No Race Diversity / No Gender Diversity condition. The key analysis would involve detecting a reliable interaction between our racial and gender diversity conditions.

If OR = 2.477 (Cohen's d = .50), expected distributions would be 14.02% in the No Race, No Gender diversity condition, but 28.77% in all other conditions. A sample size of 1060 would achieve 84.1% power to detect an interaction effect. A sample size of 1496 would achieve 95.5% power to detect an interaction effect.